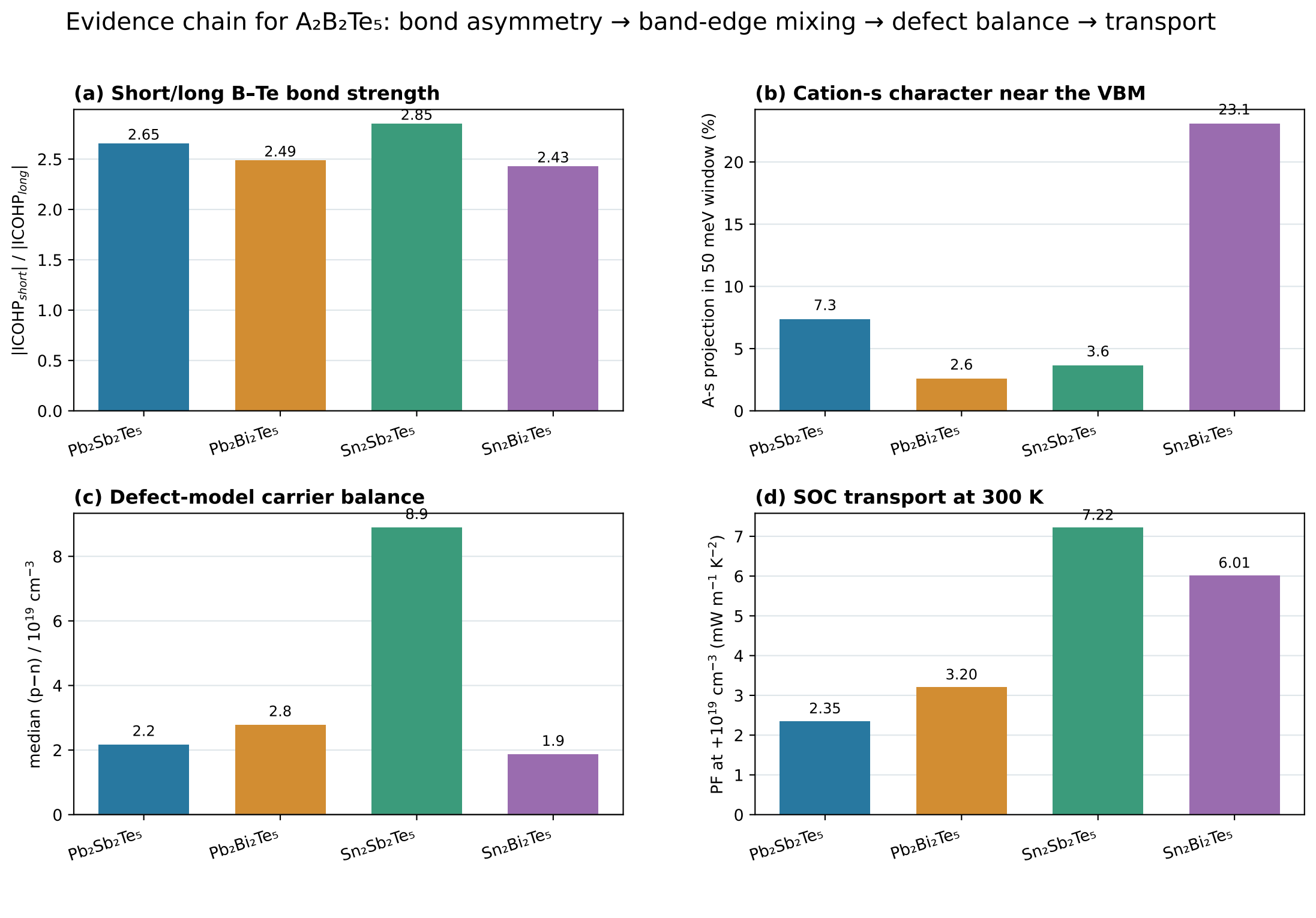
Evidence chain for A₂B₂Te₅: bond asymmetry → band-edge mixing → defect balance → transport
(a) Short/long B–Te bond strength
|ICOHP_short| / |ICOHP_long|
2.65
2.49
2.85
2.43
(b) Cation-s character near the VBM
A-s projection in 50 meV window (%)
7.3
2.6
3.6
23.1
(c) Defect-model carrier balance
median (p−n) / 10^19 cm^−3
2.2
2.8
8.9
1.9
(d) SOC transport at 300 K
PF at +10^19 cm^−3 (mW m^−1 K^−2)
2.35
3.20
7.22
6.01
Pb₂Sb₂Te₅
Pb₂Bi₂Te₅
Sn₂Sb₂Te₅
Sn₂Bi₂Te₅
Panels summarize separate calculations; they show material-dependent association, not a demonstrated causal law.